# LSSO: Solved Contextual Adaptation with Certifiable Global Mixing

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**Abstract**

Context-conditioned sequence layers commonly express adaptation through finite update rules or recurrent state evolution. We introduce LSSO, a global mixer that instead turns contextual adaptation into a solved operator: one per-sample statistic generates a compact strictly accretive system, and a direct solve returns its equilibrium. A QR soft frame lifts the compact operator to token space, while a learned scalar complement controls the unrepresented subspace. The resulting chart covers the entire open real matrix ball, including nonsymmetric and nonnormal contractions. Every realized token mixer is a strict contraction whose exact gain is computable from a matrix of size at most $r \times r$; a per-sample monotonicity margin also bounds the equilibrium, contextual sensitivity, and implicit adjoint. LSSO therefore removes unrolling, inner learning rates, and solver tolerances from the layer. We provide an algebraically equivalent native CUDA path with an analytic backward that reuses the compact factors. In a matched eight-task GenomicBenchmarks comparison, LSSO outperforms multi-head attention on seven tasks and improves unweighted macro accuracy from 75.37 to 78.53. Across four Long Range Arena tasks with sequences up to 4096 positions, LSSO attains the highest four-task average among the displayed general-purpose global mixers.

## 1 Introduction

Global interaction is central to sequence and vision encoders, yet materializing all token pairs incurs quadratic cost. Efficient alternatives compress the sequence into landmarks or latent slots, replace attention by structured recurrence, or maintain an input-selective state [28, 15, 18, 12, 9, 8, 6]. A second line of work interprets sequence processing itself as learning: fast-weight models and test-time training adapt a compact inner state to the current context [22, 23]. Together these developments suggest that the central object need not be a dense attention matrix; it can be a small state constructed from the sample.

LSSO builds on two complementary advances. XCiT shows that cross-covariance statistics can define an efficient, input-dependent global operator shared across tokens [1]. Direct Cayley parameterization provides a complete differentiable map from free coordinates to static Lipschitz-bounded network weights [27]. LSSO takes the next step: it turns static contractive weights into a compact Cayley token mixer generated anew from every input, solved exactly, and certified on the realized sample.

This raises a question for context-adaptive layers: how should that state be obtained? Finite inner optimization turns its objective, learning rate, update count, and model depth into architectural choices. ViT$^3$ shows that these choices materially shape visual test-time training and that deeper inner models can remain under-optimized [11]. MesaNet moves toward locally optimal adaptation by solving per-context regressions to a numerical tolerance, at the cost of a data-dependent iterative solve [25]. Deep equilibrium and monotone equilibrium networks instead define implicit states through root finding and monotone structure [2, 29], but still require iterative forward and backward solvers. We instead study a bidirectional mixer in which each sample defines a strongly monotone compact problem, solved once with fixed work. The goal is to retain global conditioning and linear sequence scaling while making the realized operator geometry exactly verifiable.

LSSO answers this question with a compact matrix-ball construction. Each head compresses the full sample into a shared cross-moment $Z = P^{\mathsf{T}}C$, which generates a strictly accretive matrix $K$. Its contextual state is returned by the direct solve $U^\star = (\mathbf{I} + K)^{-1}Z$. Equivalently, the compact mixer is the reflected resolvent

$$M = 2(\mathbf{I} + K)^{-1} - \mathbf{I} \tag{1}$$

of the generated accretive system. A QR soft frame $P$ and a learned scalar complement $\eta$ lift this state to tokens,

$$T = \eta \mathbf{I} + P(M - \eta \mathbf{I})P^{\mathsf{T}}. \tag{2}$$

The construction never materializes $T$. It evaluates token–rank products and one small solve, giving linear sequence scaling for fixed rank. The accretive chart is complete, its reflected resolvent spans every real strict compact contraction, and the scalar complement learns the gain on token directions outside the frame $P$.

LSSO combines three properties: noncausal global mixing, a directly solved contextual state, and an exact per-instance gain certificate. For every input, the dynamic token mixer has gain $q(X) < 1$, recoverable entirely in compact space, while a second certificate bounds the solve and its implicit adjoint. The default configuration uses the Dynamic core with Rank-Rotary and provides native CUDA inference, training forward, and analytic first-order backward paths. Compact construction and factorization share one schedule, and the backward reuses saved factors without forming an $N \times N$ token matrix or an explicit compact Cayley matrix.

Our contributions are fourfold:

1. **Solved adaptation without unrolling.** We introduce a noncausal global mixer whose sample-conditioned inner state is the exact solution of a compact accretive system. At fixed rank it scales linearly with sequence length and removes inner learning rates, update counts, and fixed-point iterations from the forward pass.

2. **The first context-conditioned Cayley token mixer.** We advance direct Cayley parameterization from static Lipschitz-bounded network weights to a compact operator generated anew for every input. The construction spans the full open real matrix ball in rank space, including nonsymmetric and nonnormal contractions.

3. **Exact per-instance certification.** For every realized token mixer, we derive its exact gain $q(X) < 1$ entirely in compact space. A complementary sample-dependent monotonicity margin $\mu > 1$ certifies the equilibrium, contextual sensitivity, implicit adjoint, and continuous-time convergence rate. Both certificates require only matrices of size at most $r \times r$ and are directly observable in trained networks.

4. **Empirical breadth and efficient execution.** In a matched three-seed comparison across eight genomic tasks, LSSO improves over MHA on seven tasks and raises macro accuracy by 3.16 points. A four-task LRA panel spans sequences up to length 4096. An algebraically equivalent native CUDA forward and analytic backward reaches up to 2.30× the measured Flash-SDPA MHA training throughput without forming a token–token map.

## 2 Related Work

**Global compact-state mixing.** Efficient Transformers replace the dense token map with projections, landmarks, kernels, or inducing states [28, 13, 4, 30]. Set Transformer uses inducing points for global set interaction [15]; Luna packs and unpacks information through a fixed-length auxiliary sequence [18]; and ABC interprets several such mechanisms as learned bounded memory [21]. Perceiver similarly mediates global exchange through latent slots [12]. LSSO advances this compact-state line by building on XCiT's central insight that an input-dependent cross-statistic can generate one compact global operator for all tokens [1]. XCiT turns token-aggregated cross-covariance into a shared channel mixer; LSSO carries the compact operator into token space, where the cross-moment generates an accretive system. Its equilibrium is solved directly, and the resulting contraction is lifted through a learned token frame without constructing a dense token matrix.

**Context as learning.** Fast-weight views interpret linear attention as programming an associative memory [22]. TTT learns an inner model from the current sequence [23], while Longhorn and test-time regression connect state updates to online least-squares objectives [17, 26]. ViT$^3$ exposes the design and optimization of inner models as a central axis in vision [11]; MesaNet moves toward more complete per-context optimization with a parallel iterative solver [25]. For bidirectional encoders, LSSO replaces both the finite update schedule and the iterative tolerance with the exact equilibrium of a per-sample strongly monotone system.

**Context-conditioned contractive geometry.** S4 and selective state-space models obtain linear sequence scaling through structured recurrence and input-dependent state evolution [9, 8, 6]. Deep equilibrium models use implicit differentiation at a root [2]; monotone operator equilibrium networks additionally obtain well-posed states from monotone structure [29]. Direct Cayley parameterization gives a complete trainable chart for static Lipschitz-bounded network weights [27]. LSSO makes this geometry contextual: every sample generates a compact accretive operator, its exact solve produces the global token mixer, and compact statistics recover the gain actually realized on that sample. Skew coordinates retain nonsymmetric and nonnormal contractions. This joins global noncausal conditioning with accretive/resolvent geometry [3] and provides a directly measurable operator guarantee in a setting where ordinary self-attention need not be globally Lipschitz [14].

**Hardware-efficient linear layers.** FlashAttention demonstrated that avoiding materialized intermediates and controlling memory traffic can matter as much as operation count [5]; Mamba likewise couples its recurrence to a hardware-aware parallel algorithm [8]. LSSO follows this algorithm–implementation co-design principle. Its native path lowers the compact algebra to fused CUDA schedules for both the forward and analytic backward. Section 6 reports its measured behavior, and Appendix C gives the lowering and workspace accounting.

# 3 LSSO

We describe one head and omit the batch index. Let $X \in \mathbb{R}^{N\times D}$ be the incoming token matrix, $d$ the head width, and $r$ the compact rank. A single input projection produces relation and content coordinates

$$[A, C] = XW_{bc} + b_{bc}, \qquad A \in \mathbb{R}^{N\times r}, \quad C \in \mathbb{R}^{N\times d}. \tag{3}$$

Heads are concatenated and passed through an ordinary output projection after mixing. The construction below is repeated independently for each head, with one compact state shared by all valid tokens in that head.

## 3.1 Masking and rank-space phase

For a boolean validity mask $m \in \{0,1\}^N$, invalid rows of both $A$ and $C$ are set to zero before any compact statistic is formed. Let

$$n = \sum_{i=1}^{N} m_i. \tag{4}$$

For a nonempty sequence, relation coordinates are normalized as $A \leftarrow A/\sqrt{n}$. An entirely masked sequence is assigned the zero output by definition. Masking before normalization makes the result invariant to appended padding.

Rank-Rotary applies a block-diagonal planar rotation to the $r$ relation coordinates before the soft frame. For centered position $p_i$ and frequencies $\{\omega_j\}_{j=1}^{r/2}$, each coordinate pair uses the angle $p_i\omega_j$. The transformation preserves each relation row's Euclidean norm and obeys the relative-phase identity

$$\mathcal{R}(p_i)^{\mathsf{T}}\mathcal{R}(p_j) = \mathcal{R}(p_j - p_i). \tag{5}$$

Rank-Rotary therefore supplies relative phase within relation rank space while preserving each row norm. Absolute or two-dimensional position embeddings remain external to the mixer.

## 3.2 QR soft frame and compact statistic

The QR soft frame maps unconstrained relation coordinates to a strict column contraction. Form the reduced QR factorization

$$\begin{bmatrix} P \\ E \end{bmatrix} R_{\mathrm{qr}} = \begin{bmatrix} A \\ \mathbf{I}_r \end{bmatrix}, \qquad \begin{bmatrix} P \\ E \end{bmatrix}^{\mathsf{T}} \begin{bmatrix} P \\ E \end{bmatrix} = \mathbf{I}_r, \tag{6}$$

where $P \in \mathbb{R}^{N\times r}$ is the token-space frame. A common positive rescaling, which leaves the exact $Q$ factor unchanged, controls the numerical range. The identity augmentation means that the factorization is defined even when $N < r$ or $A$ is rank deficient.

The frame compresses the content into one shared cross-moment,

$$Z = P^{\mathsf{T}} C \in \mathbb{R}^{r\times d}. \tag{7}$$

$Z$ aggregates all valid positions symmetrically in one noncausal pass without forming the token-level matrix in (2).

### 3.3 Accretive compact equilibrium

In the default Dynamic mode, $Z$ generates compact raw coordinates

$$\Theta = \Theta_0 + \frac{ZW_{\mathrm{drive}}}{\sqrt{n}}, \tag{8}$$

where $\Theta_0 \in \mathbb{R}^{r\times r}$ and $W_{\mathrm{drive}} \in \mathbb{R}^{d\times r}$ are learned per-head parameters. The Static ablation uses $\Theta = \Theta_0$, while Zero fixes the compact contraction to zero.

For Dynamic and Static, define

$$L(\Theta) = \mathrm{tril}(\Theta, -1) + \mathrm{Diag}\big(\mathrm{softplus}\big(\mathrm{diag}(\Theta) + c_1\big)\big)\,, \tag{9}$$
$$\Omega(\Theta) = \mathrm{triu}(\Theta, 1) - \mathrm{triu}(\Theta, 1)^{\mathsf{T}}, \tag{10}$$
$$K(\Theta) = L(\Theta)L(\Theta)^{\mathsf{T}} + \Omega(\Theta), \tag{11}$$

where $c_1 = \mathrm{softplus}^{-1}(1)$. Thus $L$ is lower triangular with positive diagonal, while $\Omega$ is skew-symmetric. The compact equilibrium and the pre-output token coordinates are

$$U^\star = (\mathbf{I}_r + K)^{-1} Z, \tag{12}$$
$$Y = \eta C + P\left[2U^\star - (1+\eta)Z\right], \qquad \eta = \tanh(\eta_{\mathrm{raw}}) \in (-1, 1). \tag{13}$$

The complement is learned independently for each head and is initialized to 0.9. In Zero, $M = 0$, which gives

$$Y = \eta(C - PZ). \tag{14}$$

At $\Theta = 0$, $L = \mathbf{I}_r$, $\Omega = 0$, $K = \mathbf{I}_r$, and the compact contraction is $M = 0$. Both nonzero-core modes start at that point; Dynamic also starts with $W_{\mathrm{drive}} = 0$.

For later use, define the reflected resolvent

$$M = 2(\mathbf{I}_r + K)^{-1} - \mathbf{I}_r = (\mathbf{I}_r - K)(\mathbf{I}_r + K)^{-1}. \tag{15}$$

Substituting $Z = P^{\mathsf{T}} C$ into (13) gives the exact token-mixing form

$$Y = TC, \qquad T = \eta \mathbf{I}_N + P(M - \eta \mathbf{I}_r)P^{\mathsf{T}}. \tag{16}$$

**Solved contextual inner state.** LSSO obtains $U^\star$ by a direct $r \times r$ solve. Section 4.3 relates this solution to a globally convergent contextual dynamics.

**Cost.** For fixed $r$ and $d$, frame formation costs $O(Nr^2)$ per head, the frame-content products cost $O(Nrd)$, and the compact solve costs $O(r^2 d + r^3)$. The mixer is therefore linear in $N$ at fixed compact dimensions and never stores an $N \times N$ token matrix. The dense input and output projections remain ordinary feature projections, as in other token mixers.

## 4 Solved and Certifiable Contextual Mixing

We analyze one realized head in exact arithmetic; mixed-precision execution is described in Section 6.

## 4.1 The QR soft frame is a strict contraction chart

**Theorem 4.1** (Soft-frame contract)**.** *Let $P$ be the upper block of the $Q$ factor in* (6)*. If $\sigma_i(A)$ are the singular values of the normalized relation matrix, then*

$$\sigma_i(P) = \frac{\sigma_i(A)}{\sqrt{1+\sigma_i(A)^2}}. \tag{17}$$

*Consequently,*

$$P^\top P \prec \mathbf{I}_r, \qquad \mathbf{I}_N - PP^\top \succ 0, \qquad 1 - \|P\|_2^2 = \frac{1}{1+\|A\|_2^2}. \tag{18}$$

*Moreover, at the level of represented frames, the construction covers every $P \in \mathbb{R}^{N\times r}$ satisfying $P^\top P \prec \mathbf{I}_r$.*

*Proof sketch.* The lower block of (6) obeys $ER_{\mathrm{qr}} = \mathbf{I}_r$. Since the augmented matrix has full column rank, $R_{\mathrm{qr}}$ and hence $E$ are nonsingular. Orthogonality of the full thin-$Q$ factor gives

$$P^\top P + E^\top E = \mathbf{I}_r. \tag{19}$$

Because $E^\top E \succ 0$, the first claim follows. Also $R_{\mathrm{qr}}^\top R_{\mathrm{qr}} = \mathbf{I}_r + A^\top A$ and $P = AR_{\mathrm{qr}}^{-1}$, which yields (17) and the exact norm margin. In particular, $\|P\|_2 < 1$, which implies $PP^\top \prec \mathbf{I}_N$ and proves the second claim. Appendix A gives the full spectral argument and constructs the inverse chart for every $P$ in the strict column ball. □

The identity augmentation supplies the strict complement $\mathbf{I}_N - PP^\top$, which allows the compact contraction to extend to every token direction. If $A_0$ denotes the masked relation matrix before division by $\sqrt{n}$, the final term in (18) becomes $1/(1+\|A_0\|_2^2/n)$. Thus bounded per-token relation energy prevents the frame margin from deteriorating merely because the sequence is longer.

**Proposition 4.2** (Length-normalized contextual drive)**.** *For the dynamic core,*

$$\|Z\|_F \le \|P\|_2 \|C\|_F, \qquad \|\Theta - \Theta_0\|_F \le \frac{\|C\|_F}{\sqrt{n}} \|W_{\mathrm{drive}}\|_2. \tag{20}$$

*The first inequality is strict relative to $\|C\|_F$ whenever $C \ne 0$. Accordingly, if $\|C\|_F^2/n \le c^2$, the magnitude of the contextual drive is at most $c\|W_{\mathrm{drive}}\|_2$, independently of sequence length.*

Under bounded average token energy, the resulting drive bound is independent of sequence length.

## 4.2 The compact core parameterizes the strict matrix ball

**Definition 4.3** (Strict accretivity)**.** *A real square matrix $K$ is strictly accretive when*

$$\mathrm{sym}(K) \coloneqq \frac{K+K^\top}{2} \succ 0. \tag{21}$$

The Cayley connection between accretive coordinates and contractions is classical in monotone operator theory [3], and direct Cayley charts have parameterized static Lipschitz-bounded network weights [27]. The result below gives the complete strict form needed by LSSO. Its role here is contextual: $K$ is generated from the current sample's cross-moment, and the resulting contraction is lifted through that sample's token frame.

**Theorem 4.4** (Accretive Cayley parameterization)**.** *The map in* (11) *ranges over all real strictly accretive $r\times r$ matrices. For every such $K$, the reflected resolvent $M$ in* (15) *is a strict spectral-norm contraction and obeys*

$$\mathbf{I}_r - M^\top M = 4(\mathbf{I}_r + K)^{-\top} LL^\top (\mathbf{I}_r + K)^{-1} \succ 0. \tag{22}$$

*Conversely, the map $K \mapsto M$ is a bijection from real strictly accretive matrices to the open real matrix ball $\{M : \|M\|_2 < 1\}$, with inverse*

$$K = (\mathbf{I}_r - M)(\mathbf{I}_r + M)^{-1}. \tag{23}$$

*Proof sketch.* The construction has $\operatorname{sym}(K) = LL^\top \succ 0$, because $L$ has positive diagonal. Conversely, for any strictly accretive $K$, the Cholesky factor of $\operatorname{sym}(K)$ and the skew remainder $K - \operatorname{sym}(K)$ recover $L$ and $\Omega$, proving completeness.

Let $Q = \mathbf{I}_r + K$. Strict accretivity implies that $Q$ is nonsingular. Using $M = (\mathbf{I}_r - K)Q^{-1}$ gives

$$\mathbf{I}_r - M^\top M = Q^{-\top}\left[Q^\top Q - (\mathbf{I}_r - K)^\top(\mathbf{I}_r - K)\right]Q^{-1} \tag{24}$$

$$= 4Q^{-\top}\operatorname{sym}(K)Q^{-1}, \tag{25}$$

which is (22). It is positive definite, so $\|M\|_2 < 1$. The inverse formula maps every strict contraction back to a strictly accretive matrix. Appendix A gives the complete converse and fractional-linear algebra. □

Because $\Omega$ is unrestricted, $K$ and $M$ may be nonsymmetric and nonnormal. The chart retains every strict contraction in rank space, and at token level, $T - T^\top = P(M - M^\top)P^\top$, so this nonsymmetric structure survives the lift.

**Theorem 4.5** (Stable solved state and implicit adjoint)**.** *Define the per-sample monotonicity margin*

$$\mu = 1 + \sigma_{\min}(L)^2 > 1, \qquad A_K = \mathbf{I}_r + K. \tag{26}$$

*Then $U^\star = A_K^{-1}Z$ exists uniquely, and*

$$\sigma_{\min}(A_K) \geq \mu, \quad \|U^\star\|_F \leq \mu^{-1}\|Z\|_F, \quad \left\|A_K^{-\top}G\right\|_F \leq \mu^{-1}\|G\|_F. \tag{27}$$

*For a second generated problem $A'_K = \mathbf{I}_r + K'$ with margin $\mu'$ and solution $U^{\star'} = A_K'^{-1}Z'$, the finite contextual perturbation obeys*

$$\left\|U^\star - U^{\star'}\right\|_F \leq \mu^{-1}\|Z - Z'\|_F + (\mu\mu')^{-1}\|K - K'\|_2\|Z'\|_F. \tag{28}$$

*Consequently, an infinitesimal perturbation obeys*

$$\|dU^\star\|_F \leq \mu^{-1}\|dZ\|_F + \mu^{-2}\|dK\|_2\|Z\|_F. \tag{29}$$

The same computable margin controls the forward state, the transposed solve used by implicit differentiation, and sensitivity to the generated compact problem. The realized-gain certificate below separately measures the contraction margin, which also depends on the skew coordinates. Appendix A gives the proof, exact implicit VJP, and a condition-number example.

## 4.3 The direct solve resolves a convergent contextual process

Let $A_K = \mathbf{I}_r + K$. The following result relates the direct solve to a continuous contextual process.

**Theorem 4.6** (Solved contextual inner process)**.** *For fixed contextual quantities $K$ and $Z$, consider the continuous dynamics*

$$\dot{U}(t) = Z - A_K U(t). \tag{30}$$

*From every initial state $U(0)$, the dynamics converge to $U^\star = A_K^{-1}Z$ and obey*

$$\|U(t) - U^\star\|_F \leq e^{-\mu t}\|U(0) - U^\star\|_F. \tag{31}$$

*The rate lower bound is independent of the skew-symmetric part of $K$.*

*Proof sketch.* Write $E(t) = U(t) - U^\star$. Since $A_K U^\star = Z$, $\dot{E} = -A_K E$. The skew part vanishes in the Frobenius inner product, so

$$\frac{d}{dt}\frac{1}{2}\|E\|_F^2 = -\langle E, A_K E\rangle_F = -\langle E, \operatorname{sym}(A_K)E\rangle_F \leq -\mu\|E\|_F^2, \tag{32}$$

because $\operatorname{sym}(A_K) = \mathbf{I}_r + LL^\top \succeq \mu\mathbf{I}_r$. Gronwall's inequality gives (31). □

Appendix A further shows that this direct solution is the infinite-step limit of stable gradient descent on the residual objective and derives its exact implicit VJP and sensitivity bound. Thus the production backward differentiates the converged state rather than a finite unroll.

## 4.4 An exact compact certificate for the realized token mixer

**Theorem 4.7** (Exact realized-gain certificate)**.** *Fix a sample and head, and let $P$, $M$, and $\eta$ be constructed as above. Write the positive eigenspace of $P^\top P$ as*

$$P^\top P = V_k S^2 V_k^\top, \qquad k = \mathrm{rank}(P), \tag{33}$$

*where $S \in \mathbb{R}^{k\times k}$ is positive diagonal, and define*

$$B = \eta \mathbf{I}_k + S V_k^\top (M - \eta \mathbf{I}_r) V_k S. \tag{34}$$

*Then the singular values of $T$ are exactly those of $B$, together with $N - k$ copies of $|\eta|$. Equivalently,*

$$q(X) := \|T\|_2 = \begin{cases} |\eta|, & k = 0, \\ \sigma_{\max}(B), & k = N, \\ \max\{\sigma_{\max}(B), |\eta|\}, & 0 < k < N, \end{cases} \qquad q(X) < 1. \tag{35}$$

*Moreover,* $\mathrm{rank}(T - \eta \mathbf{I}_N) \le r$ *and $Tv = \eta v$ for every $v \in \ker(P^\top)$.*

*Proof sketch.* Let $Q = P V_k S^{-1}$. Then $Q^\top Q = \mathbf{I}_k$, and substitution gives $T = QBQ^\top + \eta(\mathbf{I}_N - QQ^\top)$, so an orthogonal basis containing $Q$ yields the stated singular spectrum. With $\Delta = (\mathbf{I}_N - PP^\top)^{1/2}$ and $F = [P\ \Delta]$,

$$T = F\,\mathrm{Diag}(M, \eta \mathbf{I}_N) F^\top, \tag{36}$$

and $FF^\top = \mathbf{I}_N$; hence $\|T\|_2 < 1$. Finally, $T - \eta \mathbf{I}_N = P(M - \eta \mathbf{I}_r)P^\top$, which gives both the rank bound and $Tv = \eta v$ on $\ker(P^\top)$. Appendix A gives the complete spectral argument. □

The exact certificate requires only an $r \times r$ Gram eigendecomposition and a singular-value calculation of size at most $r \times r$; no $N \times N$ kernel is needed. Because the forward uses the input-dependent $T(X)$ itself, Equation (35) gives the pointwise energy statement

$$\|T(X)C(X)\|_F \le q(X)\,\|C(X)\|_F < \|C(X)\|_F \tag{37}$$

whenever $C(X) \ne 0$. The guarantee is pointwise in the input-conditioned operator. Along any fixed realized sequence of mixer maps, the gains compose: $\|T_L \cdots T_1\|_2 \le \prod_{\ell=1}^{L} q_\ell$.

**Remark 4.8** (Why the scalar complement is part of the model)**.** *Without the complement, a conventional form $\mathbf{I}_N + PAP^\top$ fixes the entire* $\ker(P^\top)$ *subspace to gain one. Equation* (36) *instead assigns that unmodeled subspace the learned per-head scalar $\eta$, while retaining the same rank-$r$ correction and the strict contraction certificate. Thus $T$ is a rank-$r$ correction to an isotropic complement.*

Appendix A proves padding isolation and noncausal permutation equivariance. The certificates characterize the contextual mixer, direct solve, and implicit adjoint; feature projections, residual paths, and MLPs retain their standard roles.

## 4.5 Certificates in trained models

We next measure how much certified margin trained models retain. Across three six-layer LRA Text checkpoints and matched held-out token-stream probes from 1K to 8K, all 9,216 head-level observations are finite and satisfy their certified conditions (Table 1). The worst realized gain is 0.9087, leaving at least 0.0913 contraction slack, and the minimum monotonicity margin is 1.1667. Solved states use at most 93.3% of their certified bound, while deterministic adjoint probes use at most 88.1%. Appendix B reports the per-layer distributions and stress-test protocol.

Table 1: Worst observation in the trained-model certificate audit. Bound usage is the measured ratio normalized by its certified upper bound $1/\mu$.

| Quantity | Certified condition | Worst observed |
|---|---|---|
| Realized gain | $q(X) < 1$ | $\max q = 0.9087$ |
| Contraction slack | $1 - q(X) > 0$ | $\min(1 - q) = 0.0913$ |
| Monotonicity margin | $\mu > 1$ | $\min \mu = 1.1667$ |
| Solved-state bound usage | $\mu \left\|U^\star\right\|_F / \left\|Z\right\|_F \leq 1$ | 0.9329 |
| Adjoint-probe bound usage | $\mu \left\|A_K^{-\mathsf{T}} G\right\|_F / \left\|G\right\|_F \leq 1$ | 0.8814 |

# 5 Sequence Classification Evidence

We evaluate LSSO as a drop-in replacement for attention in otherwise ordinary bidirectional encoders. All reported LSSO runs use the default Dynamic core, Rank-Rotary, rank 32, FP16 automatic mixed precision, and the CUDA implementation. Checkpoints are selected only by validation accuracy, with validation loss breaking an exact tie; the held-out test split is evaluated once after selection. Reported uncertainties are sample standard deviations over seeds 0, 1, and 2.

## 5.1 Matched genomic mixer comparison

GenomicBenchmarks collects binary DNA sequence-classification tasks with official train and test partitions [7]. We derive a deterministic stratified validation set only from each training partition. MHA and LSSO use the same $D = 128$, four-layer, four-head encoder, learned absolute position embeddings, MLP ratio 4, mean readout, optimizer, data order, and 40-epoch budget; only the token mixer changes. The usual physical batch is 128. Mouse Enhancers Ensembl uses physical batch 64 and two-step accumulation, preserving the same effective batch. LSSO has 33,264 fewer parameters than MHA in every task because the mixer parameterizations differ; for the 200-position tasks the totals are 786,834 and 820,098.

Table 2: GenomicBenchmarks held-out test accuracy (%), mean ± sample standard deviation over three seeds. $\Delta$ is computed from the unrounded seed means as LSSO minus MHA. Bold marks the higher mean in each row.

| Task | MHA | LSSO | $\Delta$ |
|---|---|---|---|
| Coding vs. intergenomic | $80.34 \pm 1.99$ | $\mathbf{83.23 \pm 2.46}$ | +2.89 |
| Human or worm | $65.87 \pm 0.35$ | $\mathbf{81.57 \pm 1.18}$ | +15.70 |
| Mouse enhancers Ensembl | $\mathbf{71.07 \pm 1.65}$ | $70.25 \pm 0.83$ | −0.83 |
| Human enhancers Cohn | $66.83 \pm 0.17$ | $\mathbf{67.24 \pm 0.26}$ | +0.41 |
| Human enhancers Ensembl | $76.37 \pm 0.20$ | $\mathbf{78.91 \pm 0.05}$ | +2.54 |
| Human Ensembl regulatory | $92.14 \pm 0.05$ | $\mathbf{93.10 \pm 0.02}$ | +0.96 |
| Human non-TATA promoters | $84.71 \pm 0.15$ | $\mathbf{85.41 \pm 0.17}$ | +0.70 |
| Human OCR Ensembl | $65.61 \pm 0.54$ | $\mathbf{68.55 \pm 0.31}$ | +2.93 |
| Unweighted task macro | 75.37 | **78.53** | +3.16 |

Table 2 shows that LSSO is higher on seven of eight tasks. The macro result first averages seeds within each task and then gives all eight tasks equal weight. The largest gain occurs on Human or Worm, while Mouse Enhancers Ensembl is the one negative result.

## 5.2 Long Range Arena context

Long Range Arena (LRA) tests structured reasoning and long-context classification [24]. Its reported scores are strongly affected by architectural locality: recent analysis shows that explicit local and positional structure can account for a substantial fraction of the benchmark gains [20]. Following established LRA reporting practice [9], Table 3 places our LSSO results alongside a representative set of values reported by the original benchmark and subsequent work. The divider separates general-purpose global mixers and efficient attention approximations from models centered on structured recurrence or state-space dynamics, grouping models by their primary mixing mechanism.

We evaluate LSSO on four LRA tasks using six-layer encoders, with $D = 128$ for ListOps and Retrieval and $D = 256$ for Text and Pathfinder-32. ListOps, Text, and Retrieval use learned one-dimensional absolute position embeddings with mean pooling; Pathfinder-32 uses factorized row/column embeddings and mean–maximum pooling. Their maximum sequence lengths are 2048, 4096, 4000, and 1024, respectively.

Table 3: LRA test accuracy (%) quoted from the cited sources under their reported protocols. The four-task average is unweighted over the displayed tasks. For LSSO, task scores and the per-seed four-task average are reported as mean ± sample standard deviation across three seeds. Bold marks the highest reported mean within each mechanism block.

| Model and source | ListOps | Text | Retrieval | Pathfinder | 4-task Avg. |
|---|---|---|---|---|---|
| *General-purpose global mixers and efficient-attention approximations* | | | | | |
| Transformer [24] | 36.37 | 64.27 | 57.46 | 71.40 | 57.38 |
| Linear Transformer [24] | 16.13 | **65.90** | 53.09 | 75.30 | 52.61 |
| Performer [24] | 18.01 | 65.40 | 53.82 | 77.05 | 53.57 |
| Linformer [24] | 35.70 | 53.94 | 52.27 | 76.34 | 54.56 |
| Reformer [24] | 37.27 | 56.10 | 53.40 | 68.50 | 53.82 |
| Luna-256 [18] | **37.98** | 65.78 | 79.56 | 78.55 | 65.47 |
| FNet [16] | 35.33 | 65.11 | 59.61 | 77.80 | 59.46 |
| Nyströmformer [30] | 37.15 | 65.52 | 79.56 | 70.94 | 63.29 |
| **LSSO**, three seeds | $37.60 \pm 0.65$ | $65.19 \pm 0.46$ | $\mathbf{82.88 \pm 0.08}$ | $\mathbf{78.79 \pm 0.34}$ | $\mathbf{66.11 \pm 0.24}$ |
| *Models centered on structured recurrence or state-space dynamics* | | | | | |
| S4, original [9] | 58.35 | 76.02 | 87.09 | 86.05 | 76.88 |
| S4-LegS, updated [10] | $59.60 \pm 0.07$ | $86.82 \pm 0.13$ | $90.90 \pm 0.15$ | $94.20 \pm 0.25$ | 82.88 |
| Mega [19] | **63.14** | **90.43** | **91.25** | **96.01** | **85.21** |

Across the general-purpose global-mixer block, LSSO gives the highest displayed Retrieval and Pathfinder values; it is within 0.38 points of the highest ListOps result and 0.71 points of the highest Text result. Its unweighted four-task average is also the highest in that block. More broadly, one LSSO mixer trains across symbolic, byte-level, paired-retrieval, and two-dimensional path tasks, including sequences far longer than the genomic panel. The stronger results in the second block also show the value of task-aligned structured recurrence and state-space dynamics on LRA; LSSO obtains its panel with a general-purpose global mixer.

# 6 Native CUDA Realization

The default Dynamic core with Rank-Rotary has a native, algebraically equivalent CUDA path for inference and first-order training. It avoids both the $N \times N$ token matrix and the compact Cayley matrix; compact factorization and tiled solves evaluate the equilibrium directly, while the analytic backward reuses saved factors. Mixed-precision outputs and gradients are validated against the FP64 reference. Appendix C gives the execution schedule, workspace, VJP, numerical envelope, and build coverage.

Table 4 measures the complete mixer boundary, including input and output projections. Native LSSO is faster than the canonical PyTorch implementation throughout the reported range. Against PyTorch MHA backed by Flash SDPA, native LSSO is faster from $N = 2048$ in the forward and throughout the reported $N = 1024$–$4096$ forward–backward range, reaching $1.79\times$ and $2.30\times$ speedups at $N = 4096$.

Table 4: Synchronized milliseconds per batch on an RTX 5070 Ti (lower is better), with $B = 64, D = 256, H = 8$, FP16 AMP, and $r = 32$ for LSSO. Ratios above one favor native LSSO; forward–backward includes loss and gradient reset. Bold marks the lowest latency in each row.

| $N$ | PyTorch ref. | Native | MHA | Ref./native | MHA/native |
|---|---|---|---|---|---|
| *Inference forward* | | | | | |
| 1024 | 110.216 | 2.539 | **2.131** | 43.41× | 0.84× |
| 2048 | 133.940 | **5.577** | 5.888 | 24.02× | 1.06× |
| 4096 | 169.549 | **10.135** | 18.173 | 16.73× | 1.79× |
| *Forward–backward with gradient reset* | | | | | |
| 1024 | 118.858 | **7.512** | 8.487 | 15.82× | 1.13× |
| 2048 | 153.955 | **15.260** | 22.940 | 10.09× | 1.50× |
| 4096 | 211.268 | **29.160** | 67.195 | 7.25× | 2.30× |

# 7 Limitations

The current LSSO operator is bidirectional: its compact statistic summarizes the complete valid sequence before the equilibrium is solved. It therefore does not provide causal masking, prefix-state updates, or autoregressive decoding. A causal LSSO would need to construct prefix-conditioned solved states while preserving the exact contraction certificate and linear sequence scaling; this remains the principal open extension of the method.

The reference implementation permits rank to be chosen for the workload, but the native CUDA realization is currently optimized and distributed only for the supported rank schedules in Appendix C. Extending that coverage is an engineering limitation rather than a restriction of the operator mathematics. The native path already provides complete forward and analytic backward execution for the default operator, while leaving clear systems headroom. FlashAttention-style specialization and autotuning across batch and head shapes, sequence lengths, ranks, and GPU generations offer a direct route to further wall-clock gains. The reported measurements therefore characterize the current implementation, not a performance ceiling for LSSO. The present empirical evidence covers genomic classification, four LRA tasks, and wall-clock measurements on one recent NVIDIA architecture; vision tasks and broader hardware remain unevaluated.

# 8 Conclusion

LSSO turns contextual adaptation from a finite update procedure into a solved global operator. One per-sample statistic generates a compact strictly accretive system, a direct solve returns its equilibrium, and defect completion lifts the resulting strict contraction to token space. The construction has linear sequence scaling at fixed rank and a complete native CUDA forward and analytic backward realization. GenomicBenchmarks and LRA results show that the same certifiable operator remains useful across long biological, symbolic, textual, retrieval, and spatial inputs.

# A Complete Theoretical Details

## A.1 Soft-frame spectrum and coverage

Let $A = U\Sigma V^\top$ be an SVD of the normalized relation matrix. From the positive-diagonal QR factorization in (6),

$$R_{\mathrm{qr}}^\top R_{\mathrm{qr}} = \mathbf{I}_r + A^\top A, \qquad P = AR_{\mathrm{qr}}^{-1}. \tag{38}$$

Hence

$$PP^\top = A(\mathbf{I}_r + A^\top A)^{-1}A^\top = U\,\mathrm{Diag}\left(\frac{\sigma_i(A)^2}{1+\sigma_i(A)^2}\right)U^\top. \tag{39}$$

This proves (17), including rank-deficient $A$ and $N < r$. The Cholesky/QR coordinates may differ from the symmetric-square-root form $A(\mathbf{I}_r + A^\top A)^{-1/2}$ by a right orthogonal factor, while their singular spectra and left Gram matrices agree.

Take any $P \in \mathbb{R}^{N\times r}$ with $P^\top P \prec \mathbf{I}_r$ and write $S = \mathbf{I}_r - P^\top P \succ 0$. Let $C$ be the upper-triangular Cholesky factor of $S$, so $S = C^\top C$, and set $R_{\mathrm{qr}} = C^{-1}$ and $A = PR_{\mathrm{qr}}$. Then

$$\begin{bmatrix} A \\ \mathbf{I}_r \end{bmatrix} = \begin{bmatrix} P \\ R_{\mathrm{qr}}^{-1} \end{bmatrix} R_{\mathrm{qr}}. \tag{40}$$

The left factor has orthonormal columns because $R_{\mathrm{qr}}^{-\top} R_{\mathrm{qr}}^{-1} = S$. It is therefore the unique positive-diagonal reduced QR factorization and produces the desired upper block $P$.

## A.2 Converse Cayley map and solve bound

For any $M$ with $\|M\|_2 < 1$, the matrix $\mathbf{I}_r + M$ is nonsingular. Define $K = (\mathbf{I}_r - M)(\mathbf{I}_r + M)^{-1}$. Direct substitution gives

$$K + K^\top = 2(\mathbf{I}_r + M)^{-\top}(\mathbf{I}_r - M^\top M)(\mathbf{I}_r + M)^{-1} \succ 0. \tag{41}$$

Thus $K$ is strictly accretive, and the forward and inverse fractional-linear expressions are algebraic inverses wherever defined.

Let $\mu$ be defined in (26). For a nonzero $v$, the skew part of $K$ contributes zero to the quadratic form, so

$$\langle (\mathbf{I}_r + K)v, v\rangle = \|v\|_2^2 + \left\|L^\top v\right\|_2^2 \geq \mu \|v\|_2^2 . \tag{42}$$

Cauchy–Schwarz therefore gives $\|(\mathbf{I}_r + K)v\|_2 \geq \mu \|v\|_2$, proving $\sigma_{\min}(\mathbf{I}_r + K) \geq \mu$ and $\left\|(\mathbf{I}_r + K)^{-1}\right\|_2 \leq \mu^{-1}$.

## A.3 Residual optimization, implicit derivative, and sensitivity

The solved state also uniquely minimizes

$$J(U) = \frac{1}{2}\|A_K U - Z\|_F^2, \qquad A_K = \mathbf{I}_r + K, \tag{43}$$

whose Hessian is $A_K^\top A_K \succ 0$. Gradient descent with $0 < \alpha < 2/\sigma_{\max}(A_K)^2$ obeys

$$\|U_j - U^\star\|_F \leq \rho^j \|U_0 - U^\star\|_F, \qquad \rho = \max_i |1 - \alpha\sigma_i(A_K)^2| < 1. \tag{44}$$

Hence the direct solve is both the infinite-time limit in Theorem 4.6 and the infinite-step limit of every such residual-gradient run.

Differentiating $A_K U^\star = Z$ gives

$$dU^\star = A_K^{-1}(dZ - dA_K\, U^\star). \tag{45}$$

If $G = \partial\mathcal{L}/\partial U^\star$ and $V = A_K^{-\top} G$, trace cyclicity yields the exact implicit VJP

$$\frac{\partial \mathcal{L}}{\partial Z} = V, \qquad \frac{\partial \mathcal{L}}{\partial A_K} = -VU^{\star\top}. \tag{46}$$

For two generated problems, the resolvent identity gives

$$U^\star - U^{\star\prime} = A_K^{-1}(Z - Z') + A_K^{-1}(K' - K)A_K'^{-1} Z', \tag{47}$$

which proves (28). Taking its differential gives (45) and the stated local bound. The solve and adjoint bounds above further give

$$\|U^\star\|_F \leq \mu^{-1}\|Z\|_F, \qquad \|V\|_F \leq \mu^{-1}\|G\|_F, \qquad \|dU^\star\|_F \leq \mu^{-1}\|dZ\|_F + \mu^{-2}\|dK\|_2 \|Z\|_F . \tag{48}$$

The margin $\mu$ certifies the solve and flow rate, while the realized-gain certificate separately captures the effect of skew coordinates. For example, with $L = \mathbf{I}_2$ and $\Omega = a\left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$, $\mu = 2$ for every $a$, while the associated $\|M\|_2$ approaches one as $|a| \to \infty$. This is why the realized token gain is computed separately by (35).

### A.4 Exact compact token spectrum

With the notation of Theorem 4.7, $Q = PV_kS^{-1}$ has orthonormal columns. Substituting $P = QSV_k^{\mathsf{T}}$ into (16) gives

$$T = QBQ^{\mathsf{T}} + \eta(\mathbf{I}_N - QQ^{\mathsf{T}}). \tag{49}$$

If $Q_\perp$ completes $Q$ to an orthogonal matrix, then for $k > 0$

$$\begin{bmatrix} Q & Q_\perp \end{bmatrix}^{\mathsf{T}} T \begin{bmatrix} Q & Q_\perp \end{bmatrix} = \begin{bmatrix} B & 0 \\ 0 & \eta\mathbf{I}_{N-k} \end{bmatrix}. \tag{50}$$

This proves the exact singular-spectrum union and the nonzero-rank cases of (35). The case distinction matters when $N < r$ and $P$ has full row rank: then $k = N$, there is no token-space complement, and $|\eta|$ contributes no complement singular value. Since $B$ may be nonnormal, its largest singular value, not its spectral radius, is the certificate. If $k = 0$, then $P = 0$ and $T = \eta\mathbf{I}_N$, giving the first case directly.

### A.5 Padding isolation and permutation equivariance

If an invalid row has $A_i = C_i = 0$, the upper QR block has $P_i = 0$, so Equation (13) gives $Y_i = 0$. Explicit output masking preserves this value after the feature projection.

Let $\Pi$ permute token rows together with their mask and position identifiers. All rowwise projections, masking, normalization, and phase rotations commute with $\Pi$. Uniqueness of the positive-diagonal reduced QR factorization gives $P' = \Pi P$, while $Z' = P'^{\mathsf{T}}C' = P^{\mathsf{T}}C = Z$. The generated $K$, $M$, and $\eta$ are unchanged, and Equation (16) yields

$$\mathrm{LSSO}(\Pi X, \Pi m, \Pi p) = \Pi\,\mathrm{LSSO}(X, m, p). \tag{51}$$

## B Certificate Diagnostic Protocol

We use the validation-selected seed-0, seed-1, and seed-2 LRA Text checkpoints. Each is a six-layer, eight-head, rank-32 LSSO encoder trained at maximum length 4096. Sixteen probes are formed by concatenating the held-out validation token stream and taking matched prefixes of length 1024, 2048, 4096, and 8192. Thus a probe at one length is a prefix of the same probe at every longer length. This produces $3 \times 16 \times 6 \times 8 \times 4 = 9{,}216$ head-level observations.

At each layer, diagnostics are evaluated in FP64 from the realized compact factors. The exact gain $q(X)$ is obtained from the compact certificate block in Equation (34); no $N \times N$ operator is formed. We record the monotonicity margin $\mu$, the normalized solved-state ratio $\|U^\star\|_F \,/\, \|Z\|_F$, and the normalized adjoint ratio

$$\frac{\left\|A_K^{-\mathsf{T}}G\right\|_F}{\|G\|_F}, \qquad A_K = \mathbf{I} + K. \tag{52}$$

Here $G$ is a deterministic unit-Frobenius Rademacher matrix, shared across lengths for each probe–layer pair, that provides a reproducible probe of the implicit adjoint. Both ratios are checked against their shared upper bound $1/\mu$; equivalently, $\mu\,\|U^\star\|_F \,/\, \|Z\|_F \le 1$ and $\mu\,\left\|A_K^{-\mathsf{T}}G\right\|_F \,/\, \|G\|_F \le 1$ are required for every observation.

We treat 8K as an operator-level stress test. The learned absolute position table is repeated modulo its 4096-position training range, while Rank-Rotary receives centered positions over the full probe. This protocol tests the mixer beyond its training length without introducing a new learned-position extrapolation rule. The diagnostic runner writes raw JSONL observations, a CSV summary, protocol metadata, and the plotted panel.

Figure 1 shows the corresponding depth and length distributions. The realized-gain curves coincide because the learned scalar complement $|\eta|$ dominates Equation (35) in these checkpoints, yielding a stable worst-case gain across probe lengths. The compact-core diagnostics remain similarly stable: the monotonicity margin is preserved, the solved-state ratio stays bounded at 8K, and the adjoint-probe ratio remains near 0.50 across depth.

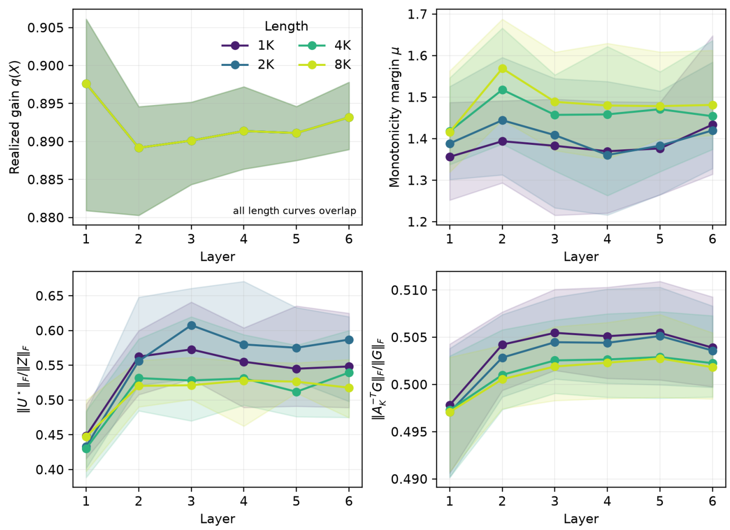


Figure 1: Per-layer certificate diagnostics from three trained LRA Text checkpoints. Curves show medians and shaded 10th–90th percentiles over held-out token-stream probes and heads. All four realized-gain curves overlap; the 8K points are operator-level stress measurements.

# C CUDA Implementation Details

## C.1 Execution contract

The native backend implements the complete default operator, Dynamic with Rank-Rotary, for inference and first-order training. It is algebraically equivalent to the equations in Section 3. The Static, Zero, and Rank-Rotary-off ablations remain reference-only and are rejected at the native boundary. Unsupported inputs never silently fall back to another implementation, and higher-order gradients are outside the native contract.

The public CUDA entry point accepts FP16 or FP32 activations, while learned parameters remain FP32. The joint input projection in Equation (3) is evaluated outside the native mixer with FP32 operands and output, with TF32 enabled on supporting hardware. Its contiguous FP32 relation and content coordinates are passed to the native operator. The ordinary output projection is also outside the native boundary; it uses FP16 Tensor-Core operands with FP32 accumulation before the result is cast back to the activation dtype. The implementation therefore covers the complete nonlinear mixer and its first-order derivative; the surrounding feature projections remain separate kernels.

The schedule supports

$$r \in \{16, 32, 48, 64\} \tag{53}$$

and any positive head width $d$ within ordinary allocation and launch limits. Rank and head width are independent, including $r > d$. Runtime right-hand-side panels have width 32 with zero-filled tails, so $d$ need not belong to a fixed set. BF16 is not part of this numerical contract.

Boolean validity masks have shape $[B, N]$, and position identifiers have shape $[N]$ or $[B, N]$. Invalid projected rows are zeroed before every compact statistic, all sample-dependent normalizations use $\sqrt{n_{\text{valid}}}$, and invalid output rows are zeroed again. This covers gapped padding and entirely masked samples in one generic schedule. Position identifiers and valid-token counts are metadata rather than differentiable inputs.

## C.2 Lowering to tiled kernels

The forward pass begins with a rank-phase schedule that constructs the Rank-Rotary sine and cosine table. Consecutive default positions are cached by device, rank, and sequence length; explicit positions use an invocation-local table. One relation-input kernel combines phase rotation, $1/\sqrt{n_{\text{valid}}}$ normalization, detached common scaling, and storage of the scaled relation coordinates.

The soft frame uses the Gram–Cholesky form algebraically equivalent to the positive-diagonal reduced QR in Equation (6). Short sequences use a single-block FP32 Gram reduction. At 32 or more 32-token tiles, independent partial Grams are reduced in ascending token order before the same Cholesky factorization. Tiled triangular solves then materialize the frame. The cross-moment $Z = P^{\mathsf{T}}C$ is produced in independent token and right-hand-side tiles using FP16 Tensor-Core multiplicands and FP32 accumulation; no more than 64 token partials are live at once.

A compact-core kernel evaluates

$$\Theta = \Theta_0 + \frac{ZW_{\mathrm{drive}}}{\sqrt{n_{\mathrm{valid}}}}, \tag{54}$$

constructs $L$ and $\Omega$, forms $LL^{\mathsf{T}}$ using IEEE FP32 FMA, assembles $\mathbf{I} + K$, and performs partial-pivot LU factorization. Separate 32-column FP32 solve panels compute $U^\star = (\mathbf{I} + K)^{-1}Z$. The final tiled kernel directly evaluates

$$Y = \eta C + P[2U^\star - (1+\eta)Z]. \tag{55}$$

Thus neither the Cayley matrix $M$ nor a token-space transition matrix is stored.

## C.3 Analytic backward and activation storage

Training records a private contiguous FP32 tape per sample and head together with one-based INT32 LU pivots. The backward reuses the saved LU factors to apply the transposed equilibrium solve, then evaluates analytic VJPs for the dynamic core, accretive parameterization, compact cross-moment, soft frame, and Rank-Rotary transform. It does not differentiate through an unrolled solver. The token-level frame and relation derivatives are fused, so the single-consumer frame adjoint remains on chip rather than making a global $[BH, N, r]$ round trip. The complement reduction reconstructs its frame–content term in FP32 within an existing compact kernel. Buffers are reused after their last consumer: the compact-core matrix adjoint becomes the frame adjoint, and compact-state partial storage is overwritten after its reduction.

Let $S = BH$, $T_{32} = \lceil N/32 \rceil$, $T_{64} = \lceil N/64 \rceil$, $Q = \lceil d/32 \rceil$, and $C_T = \min(T_{32}, 64)$. Excluding packed input, output, returned gradients, Rank-Rotary phase tables, and allocator bookkeeping, the saved training tape contains

$$W_{\mathrm{train}} = S(2Nr + 3r^2 + 2rd + 1) \tag{56}$$

FP32 values and $Sr$ INT32 pivots. The compact inference workspace contains

$$W_{\mathrm{infer}} = S(Nr + 2r^2 + 2rd + 1) \tag{57}$$

FP32 values, plus $Sr$ transient INT32 pivots. Rank-Rotary phase storage is additional: a shared table contains $Nr$ FP32 values, while per-sample positions require $BNr$. Default consecutive-position tables live in a bounded persistent cache; explicit position tables are invocation-local.

The long-sequence Gram schedule has the logical live storage

$$W_{\mathrm{Gram}} = ST_{32}\frac{r(r+1)}{2} \tag{58}$$

in FP32. Tape-producing calls with $r \in \{16, 32, 48\}$ place these partials in the not-yet-materialized frame region; inference and $r = 64$ training allocate them separately. The cross-moment producer uses

$$W_{\mathrm{cross}} = SQC_T r\, 32 \tag{59}$$

FP32 values, and its lifetime does not overlap the Gram temporary. Native backward scratch contains

$$W_{\mathrm{backward}} = S(T_{64}rd + T_{64} + 3rd + 2r^2) \tag{60}$$

FP32 values. These formulas describe the principal tape/workspace and the explicitly listed scratch tensors owned by the native mixer; full-module peak memory additionally includes phase storage, feature projections, external autograd state, returned gradients, and allocator caching.

## C.4 Numerical and build coverage

The backend follows a fixed mixed TF32/FP16-Tensor-Core/FP32 contract. The joint input projection uses TF32 where supported. Selected compact contractions and the readout use FP16 multiplicands with FP32 accumulation. Rank-Rotary phases, the soft-frame Gram and Cholesky factorization, the one-ULP-interiorized complement, $LL^{\mathsf{T}}$, LU factorization, and triangular solves remain FP32. This fixed precision partition applies to every supported configuration.

FP64 evaluation of the sole reference operator is the numerical oracle. The CUDA acceptance envelope requires finite outputs and gradients, forward relative $\ell_2$ error at most $5 \times 10^{-3}$, and input- and parameter-gradient relative $\ell_2$ error at most $10^{-2}$. The oracle suite covers all four ranks, non-tile-aligned head widths, lengths through 4776, the $N = 4096, D = 256, H = 8, r = 32$ long-sequence setting, parallel Gram reduction, batch-specific positions, gapped masks, entirely masked samples, and agreement between inference and tape-producing forward entry points. Acceptance is defined by these mixed-precision error envelopes.

CUDA 12.8 with cuBLASDx and cuSolverDx from MathDx 25.12.1 builds separate artifacts for

$$\mathrm{SM} \in \{75, 80, 86, 87, 89, 90, 100, 120\}. \tag{61}$$

Each loaded binary verifies its compiled SM and native contract version, and release metadata records the PyTorch version, CUDA runtime, C++ ABI, and artifact hashes. Physical latency and numerical measurements in this paper use the RTX 5070 Ti specified below.

## C.5 Wall-clock protocol

Table 4 was measured under WSL2 on one NVIDIA GeForce RTX 5070 Ti (16 GB, SM120), driver 610.88, PyTorch 2.11.0+cu128, CUDA 12.8, and native contract version 6. Training measurements use training mode, while inference measurements use evaluation mode. Each process pregenerates a contiguous input tensor and enables TF32. LSSO uses the default DYNAMIC core with Rank-Rotary, consecutive positions, and no padding mask. The comparison is PyTorch `nn.MultiheadAttention` with bias, zero dropout, batch-first layout, and `need_weights=False`, using PyTorch's Flash SDPA forward and backward kernels.

The timed boundary includes both operators' input and output projections, but excludes residual connections, MLPs, data loading, and optimizer updates. The forward–backward measurement differentiates a squared-mean scalar loss and includes setting the input gradient to `None` every iteration and model parameter gradients to `None` after each iteration. No CUDA Graph or `torch.compile` path is enabled. Every configuration runs 50 untimed warmup iterations, followed by seven repetitions of 100 iterations. Each repetition is bracketed by `torch.cuda.synchronize()`, and the reported number is the median synchronized host wall-clock time per iteration. Batch, width, head count, rank, precision, and statistical protocol are fixed across all sequence lengths and implementations. The benchmark entry point is `benchmarks/benchmark_ball.py`.

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